dense3_claude: an integrality reduction for the corner method, small dense patterns, and $\mathcal{K}_5 = 3$

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Abstract

We work in the setting and notation of the paper `q14paper_claude/main.tex` ("the paper"). The paper's only tool that beats the counting barrier of Remark rem:nearlimits(3) for 2-sparse families is the *corner method* of Theorem thm:H (the fractional point of HMMS followed by a finite case analysis), which it runs at threshold $c = 2$ for up to five functions. We observe that by Lemma lem:cover it suffices to run the corner method at any threshold $c < 3$ (more generally $c < D$ to prove $D - 1$): if some instance with given supports has discrepancy larger than $D - 1$, some instance with the same supports has discrepancy at least D. At $c = 2.99$ the necessary conditions are much stronger; empirically, dense cases that our runs at $c = 2$ did not finish become easy. With exact-arithmetic branch-and-bound we obtain (computer-assisted): (i) every 2-sparse family of at most six functions has discrepancy at most 2, in particular K_6, $K_{3,3}$ and the octahedron with arbitrary bundle sizes, which answers the paper's question about K_6 (Open Problem 2); (ii) the leveled conjecture of Section sec:leveled holds for all graphs on at most five vertices, and for six-vertex graphs with at most two vertices of level 1; consequently every 2-sparse family of eight functions has discrepancy at most 3 (the paper: 4), and every seven-vertex pattern other than K_7 minus a subgraph of C_4, $3K_2$ or K_3 carries only families of discrepancy at most 2; (iii) $\mathcal{K}_5 = 3$, which answers Open Problem 5 of the paper, $\mathcal{K}_6 = 4$, and $\mathcal{K}_8 \geq 5$. Separately, a weighted union bound in the randomized rounding of HMMS gives disc $\leq \sqrt{2\ln(2t)\, n}$ for every t-sparse family of n functions; for $t = 2$ this is $< 1.666\sqrt{n}$ for arbitrary bundle sizes (e.g. K_n), removing the logarithm and the dependence on bundle multiplicities. Further consequences: blocks with at most six vertices, graphs whose 3-core has at most six vertices. Runs for seven and eight 2-sparse functions and for $\mathcal{K}_7$ are reported in Section 6 with their exact status; their outcomes are not claimed. We also record precise obstructions: on support structures containing a bundle of at least 25 items (resp. of at least n items), any argument that uses only the HMMS fractional point, the sparsity pattern of the fractional items, and mean-zero Lipschitz residuals (in particular the corner method) cannot prove discrepancy 2 for $n \geq 25$ functions (resp. cannot give $o(\sqrt{n})$); it also cannot prove the leveled conjecture for six vertices with three level-1 vertices. We describe, heuristically, why parallel (multi-hub) transfers of product form reduce to sequential cheap deletions. None of this bounds the discrepancy of K_n by a constant.

1 Setting

We use the paper's definitions: instances $(g_v, U_v)_{v \in V}$ with g_v odd and 2-Lipschitz on $\{\pm 1\}^{[m]}$ depending only on U_v; 2-sparse supports; disc; c-coarse sets; leveled graphs (G, ℓ) and their goodness (Section sec:leveled); the multilinear extension G_v and the HMMS fractional point (Corollary cor:lemma9). For a finite set J, $p \in [0,1]^J$ and $X \sim w(p)$ we write $w(p)$ for the product distribution on $\{\pm 1\}^J$ with $\Pr[X_k = 1] = p_k$.

2 The integrality reduction

Lemma 2.1. *Let $(U_v)_{v\in V}$ be supports on $[m]$ and let $D_v \geq 1$ be integers. Suppose that some instance $(g_v, U_v)_{v\in V}$ has the property that for every $x \in \{\pm1\}^{[m]}$ some v satisfies $|g_v(x)| > D_v - 1$. Then some instance $(g'_v, U_v)_{v\in V}$ with the same supports has the property that for every x some v satisfies $|g'_v(x)| \geq D_v$.*

Proof. This is the leveled form of Lemma lem:cover of the paper; we include the proof. Let $P_v = \{z \in \{\pm1\}^{U_v} : g_v(z) > D_v - 1\}$. For $z, z' \in P_v$, $2\operatorname{dist}(z, -z') \geq g_v(z) - g_v(-z') = g_v(z) + g_v(z') > 2D_v - 2$, and since distances are integers, $\operatorname{dist}(z, -z') \geq D_v$. Put $g'_v = \operatorname{dist}(\cdot, -P_v) - \operatorname{dist}(\cdot, P_v)$ on $\{\pm1\}^{U_v}$ (and $g'_v \equiv 0$ if $P_v = \emptyset$). Distance functions are 1-Lipschitz, so g'_v is 2-Lipschitz; it is odd because $\operatorname{dist}(-z, -P_v) = \operatorname{dist}(z, P_v)$; it depends only on U_v. On P_v we have $\operatorname{dist}(\cdot, P_v) = 0$ and $\operatorname{dist}(\cdot, -P_v) \geq D_v$, so $g'_v \geq D_v$; by oddness $g'_v \leq -D_v$ on $-P_v$. For every x some v has $|g_v(x)| > D_v - 1$, i.e. $x|_{U_v} \in P_v \cup (-P_v)$, and then $|g'_v(x)| \geq D_v$. □

Corollary 2.2. *Let $\mathcal{S}$ be a set of support structures and let $D_v \geq 1$ be integers and $c_v \in [D_v - 1, D_v)$ reals. If every instance whose supports lie in $\mathcal{S}$ has a colouring with $|g_v| \leq c_v$ for all v, then every such instance has a colouring with $|g_v| \leq D_v - 1$ for all v. In particular: if every instance with supports in $\mathcal{S}$ has discrepancy at most 2.99, then at most 2; if every family of n functions has discrepancy at most $D - 0.01$, then $\mathcal{K}_n \leq D - 1$.*

Proof. Otherwise some instance with supports in $\mathcal{S}$ has, at every x, some v with $|g_v(x)| > D_v - 1$; Lemma 2.1 gives an instance with the same supports in which every x has some $|g'_v(x)| \geq D_v > c_v$, contradicting the hypothesis. □

Lemma 2.3 (leveled form of Lemma lem:good)**.** *A leveled graph (G, ℓ) is good if and only if every 2-sparse instance $(g_v, U_v)_{v\in V}$ whose graph H is a subgraph of G has a colouring with $|g_v| \leq \ell_v$ for all v.*

Proof. As for Lemma lem:good. If (G, ℓ) is good and (g_v, U_v) is such an instance, the sets $\{|g_v| \leq \ell_v\}$ are ℓ_v-coarse, and a solution is the required colouring. Conversely, given a leveled constraint system with $Z_v \supseteq \{|\varphi_v| \leq \ell_v\}$, the instance (φ_v, U_v) has the same supports, and a colouring with $|\varphi_v| \leq \ell_v$ for all v is a solution. □

In the applications of Corollary 2.2 below, $\mathcal{S}$ is the set of 2-sparse support structures on a vertex set V whose graph is a subgraph of a fixed graph G (for Lemma 2.3), or all support structures with n functions (for $\mathcal{K}_n$).

The reduction is useless for the deletion method (a c-coarse set is $\lfloor c \rfloor$-coarse, since Hamming distances are integers, Lemma lem:coarse), but it is very effective for the corner method, whose conditions are averages and hence genuinely real-valued.

3 The corner method with real thresholds

Definition 3.1. A *pattern* on j items for a function set V is a family $(J_v)_{v\in V}$ of subsets of $[j]$. It is *2-sparse* if every $k \in [j]$ lies in one or two of the J_v. Given thresholds $c_v > 0$, a point $p \in [0,1]^j$ admits a *bad configuration* if there are 2-Lipschitz functions $h_v \colon \{\pm1\}^{J_v} \to \mathbb{R}$ with $\mathbb{E}_{w(p)} h_v(X|_{J_v}) = 0$ such that every $z \in \{\pm1\}^{[j]}$ has some v with $|h_v(z|_{J_v})| > c_v$.

Lemma 3.2 (reduction; the proof of Theorem thm:H of the paper)**.** *Let $(U_v)_{v\in V}$ be supports, $|V| = n$, and $c_v > 0$. Suppose that for every $j \leq n$, every pattern $J_v = \{k : e_k \in U_v\}$ obtained from*

j distinct items $e_1, \dots, e_j$ that lie in at least one support, and every $p \in (0,1)^j$, no bad configuration exists. Then every instance with supports (U_v) has a colouring with $|g_v| \le c_v$ for all v.

Proof. By Corollary cor:lemma9 there is $x \in [-1,1]^m$ with at most n fractional coordinates at which all multilinear extensions vanish. Fractional coordinates in no support are rounded arbitrarily; let $e_1, \dots, e_j$ ($j \le n$) be the others and $p_k = (1 + x_{e_k})/2 \in (0,1)$. Fix the integral coordinates; the restriction h_v of g_v depends only on the fractional items in U_v, is 2-Lipschitz, and has $\mathbb{E}_{w(p)} h_v = G_v(x) = 0$. If no corner z had $|h_v(z)| \le c_v$ for all v, the h_v would be a bad configuration. A good corner, completed by the integral coordinates, is the required colouring. □

For 2-sparse supports the patterns are 2-sparse. For arbitrary families of n functions it suffices to treat the *full* pattern $J_v = [j]$ for all v, because a function on $\{\pm 1\}^{J_v}$ is also a function on $\{\pm 1\}^{[j]}$ with the same Lipschitz constant and mean.

Lemma 3.3 (necessary conditions; Lemma lem:necessaryc of the paper)**.** *If p admits a bad configuration, then there are sets $U_v^+, U_v^- \subseteq \{\pm 1\}^{J_v}$ with*

(C1) $\operatorname{dist}(U_v^+, U_v^-) \ge \lfloor c_v \rfloor + 1$*;*

(C2) $\mathbb{E}_{w(p)} \operatorname{dist}(X|_{J_v}, U) > c_v/2$ *for each nonempty* $U \in \{U_v^+, U_v^-\}$*;*

(C3) every $z \in \{\pm 1\}^{[j]}$ *has some* v *with* $z|_{J_v} \in U_v^+ \cup U_v^-$*.*

Conditions (C1) and (C2) are inherited by subsets.

Proof. $U_v^{\pm} = \{\pm h_v > c_v\}$. (C3) is the definition. For $u \in U_v^+$ and all z, $h_v(z) \ge h_v(u) - 2\operatorname{dist}(z,u) > c_v - 2\operatorname{dist}(z,u)$, hence $h_v(z) > c_v - 2\operatorname{dist}(z, U_v^+)$; taking expectations, $0 > c_v - 2\mathbb{E}\operatorname{dist}(X|_{J_v}, U_v^+)$. Similarly for U_v^-. For $u \in U_v^+$, $u' \in U_v^-$, $2\operatorname{dist}(u,u') \ge h_v(u) - h_v(u') > 2c_v$, and the distance is an integer. Shrinking a set can only increase the distances in (C1) and (C2). □

Normalisations. (N1) Negating coordinate k in all functions replaces p_k by $1 - p_k$ and preserves the pattern and all hypotheses, so we may assume $p \in [0, \frac{1}{2}]^j$. (N2) If items k and k' lie in exactly the same sets J_v, exchanging them preserves the pattern, the thresholds and all hypotheses; so we may assume $p_k \le p_{k'}$ for $k < k'$ in each class of such items. (N3) Replacing h_v by $-h_v$ exchanges U_v^+ and U_v^-; if $J_v = J_{v'}$ and $c_v = c_{v'}$, exchanging h_v and $h_{v'}$ preserves everything.

Box relaxation. For a set U, $p \mapsto \mathbb{E}_{w(p)} \operatorname{dist}(X|_{J_v}, U)$ is multilinear, so its maximum over a box $\prod_k [a_k, b_k]$ is attained at a vertex. Call a pair (U_v^+, U_v^-) *admissible on the box* if it satisfies (C1) and, for each nonempty part, (C2) at some vertex of the box (the vertices for the two parts may differ). If some p in the box admits a bad configuration, there is a cover (C3) by pairs admissible on the box. A box is *cleared* if no such cover exists; then no point of it admits a bad configuration.

Remark 3.4 (exactness at a point)**.** At a fixed p the conditions are also sufficient: if (U_v^+, U_v^-) satisfy (C1), (C2) and (C3), then p admits a bad configuration. Indeed, for $U_v^+ \ne \emptyset$ let $h^{\mathrm{lo}} = c_v + \varepsilon - 2\operatorname{dist}(\cdot, U_v^+)$ and, for $U_v^- \ne \emptyset$, $h^{\mathrm{hi}} = -c_v - \varepsilon + 2\operatorname{dist}(\cdot, U_v^-)$, with $\varepsilon > 0$ so small that (C2) still holds with $c_v + \varepsilon$ and $c_v + \varepsilon < \lfloor c_v \rfloor + 1$. Both are 2-Lipschitz, and by (C1) both are $> c_v$ on U_v^+ and $< -c_v$ on U_v^-; hence so is every convex combination $h_t = (1-t)h^{\mathrm{lo}} + t h^{\mathrm{hi}}$. By (C2), $\mathbb{E}h_0 < 0 < \mathbb{E}h_1$, so some h_t has mean 0. (If $U_v^- = \emptyset$ use $h^{\mathrm{lo}} + s$, $s \ge 0$; if both parts are empty use 0.) Thus the only relaxation in the method is the box relaxation, and a cover found at a single point is a genuine bad configuration.

The search. For each box a depth-first search looks for a cover: it picks an uncovered corner with the fewest options (pairs $(v, \pm)$ for which adding the local pattern keeps the pair admissible), and branches over these options. Completeness. Suppose some cover by admissible pairs exists. We show by induction along one branch that the following invariant holds at every node: *some cover C^* by admissible pairs contains the current partial sets* (part by part). It holds at the root. At a node with current partial sets contained in C^*, the chosen uncovered corner x is covered in C^* by some (v, s), i.e. $x|_{J_v} \in C^*_{v,s}$. If v is not yet used on the branch, we may first modify C^*: exchange the two parts of v (legitimate by (N3), and harmless since v has no points in the current partial sets) so that $s = +$; and if a lower-indexed function v' with $J_{v'} = J_v$ and $c_{v'} = c_v$ is also unused, exchange the pairs of v and v' in C^* (again legitimate by (N3) and harmless, both being unused). After these modifications C^* is still an admissible cover containing the current partial sets, and the label (v, s) is among the labels the search tries at x; adding $x|_{J_v}$ to the partial part (v, s) keeps it inside C^*, and the new partial pair is admissible because subsets of admissible pairs are admissible. So the invariant propagates; since every step covers a new corner, the branch ends with a cover, which the search finds. Boxes that are not cleared are bisected along the longest side, down to a fixed depth; the root box is $[0, \frac{1}{2}]^j$, and boxes disjoint from the region of (N2) are skipped (only consecutive items with the same relevant set are compared, which is sound).

Arithmetic. Box endpoints are dyadic with a common denominator 2^E, and all weights $\prod_k (p_k \text{ or } 1 - p_k)$ at box vertices are integers over $2^{E|J_v|}$; all sums and comparisons are exact in 128-bit integers. The quantities compared are $2c_{\text{den}}S$ and $c_{\text{num}}2^{E|J_v|}$, where $c_v = c_{\text{num}}/c_{\text{den}}$ and $S \le |J_v|\, 2^{|J_v|}\, 2^{E|J_v|}$; in all runs $c_{\text{den}} = 100$, $c_{\text{num}} \le 499$ and $|J_v| \le 8$, so both are below $2^{E|J_v|+20}$, and the program refuses to run when $E \cdot \max_v |J_v| + 20 > 126$. This guard fired only for $j = 1$ in runs with a large depth parameter; those cases ($j = 1$ is trivial by Lemma lem:onefrac) were rerun with a small depth. The program also stops if a bisection would leave the dyadic grid (this never happened). The variant used for $n = 8$ uses 256-bit corner sets but the same 128-bit arithmetic.

4 Results

Theorem 4.1 (computer-assisted)**.** *Every 2-sparse instance (in particular every 2-sparse family of 1-Lipschitz functions, with arbitrary bundle sizes and private items) with at most six functions has discrepancy at most 2.*

Proof. By Corollary 2.2 (supports with at most six functions, $D_v = 3$, $c_v = 2.99$) it suffices to prove discrepancy at most 2.99, and by Lemma 3.2 it suffices to show that no 2-sparse pattern with $j \le 6$ items on six functions admits a bad configuration at threshold 2.99 for any $p \in (0, 1)^j$. Patterns with fewer functions are patterns on six functions in which some J_v are empty. Up to relabelling the functions there are $2, 7, 23, 76, 245, 782$ such patterns for $j = 1, \ldots, 6$ (enumerated twice, by brute-force canonisation and by augmentation, with the same lists; the sum of the orbit sizes equals the number $\binom{21+j-1}{j}$ of labelled patterns in each case). For every pattern the branch-and-bound above clears all of $[0, \frac{1}{2}]^j$: in total 391 boxes for $j \le 5$ and 1256 boxes for $j = 6$, with no uncleared box. The computation was done twice, by our C program (rerun with the final source: `logs_claude/final/six_j*.log`) and by the paper's own Python checker `cover_bnb.py` (run verbatim on our pattern lists, without the sorting (N2): `final/paperchecker_n6_j1to5.log`, `indep/j6_*.log`), with identical box counts for every pattern file; the referee's independent exact checker reproduced the counts for $j \le 5$ and part of $j = 6$. □

Corollary 4.2. *For 2-sparse families with arbitrary bundle sizes and private items:*

(a) *if* $H(\mathcal{F})$ *has at most six vertices (e.g.* K_6, $K_{3,3}$, *the octahedron* $K_{2,2,2}$*), then* $\operatorname{disc}(\mathcal{F}) \le 2$, *which is tight;*

(b) *if every block of* $H(\mathcal{F})$ *has at most six vertices, then* $\operatorname{disc}(\mathcal{F}) \le 2$*;*

(c) *if repeatedly deleting vertices with at most two neighbours reduces* $H(\mathcal{F})$ *to a graph on at most six vertices (i.e. the* 3*-core of* $H(\mathcal{F})$ *has at most six vertices), then* $\operatorname{disc}(\mathcal{F}) \le 2$*;*

(d) *the bound* $\lfloor n/2 \rfloor$ *of the paper for* n *functions becomes* 2 *for* $n = 6$*, and* 3 *for* $n = 8$ *(Corollary 4.6, which uses Theorem 4.4); the schedule search* `kn_schedule_bases_claude.py` *with the proved bases gives no further improvement for* $n \le 20$*.*

Proof. (a) Lemma lem:good and Theorem 4.1; tightness as in Corollary cor:leveled(i) of the paper. (b) Corollary cor:onesum. (c) Order the vertices in the deletion order, followed by the core. In the leveled framework with all levels 2, delete the vertices in this order by Lemma lem:cheap with J equal to all current neighbours ($|J| \le 2 = \ell_h$), which changes no level; the remaining core with all levels 2 is good by Theorem 4.1 and Lemma lem:good. (d) is (a). □

Theorem 4.3 (computer-assisted)**.** *Every* 2*-sparse instance with at most seven functions in which no two functions share seven or more items has discrepancy at most* 2*. In particular every* 2*-sparse family on* K_7 *with bundles of size at most* 6 *(e.g. the simple* K_7*, for which the SAT search of the paper did not finish) has discrepancy at most* 2*.*

Proof. As for Theorem 4.1, with seven functions. Up to relabelling there are $2, 7, 23, 78, 264, 915, 3111$ patterns with $j = 1, \dots, 7$ items (orbit sums checked). Exactly one of them has two functions sharing seven fractional items (the pattern "$\{0,1\}^{\times 7}$"), and under the hypothesis it does not arise in Lemma 3.2, because the fractional items are distinct items of the instance and every bundle has at most six. All other 4399 patterns are cleared at $c = 2.99$: 1801 boxes for $j \le 6$ and 7912 boxes for the 3110 patterns with $j = 7$ (runs with v3 and v7, `logs_claude/n7c299/`, `n7c299v7/`); the paper's checker confirms all patterns with $j \le 6$ and 1878 of the 3110 with $j = 7$ with identical box counts (`indep7/`); the rerun with the final source (`final/seven_j*.log`) reproduces all of them, with 7912 boxes for $j = 7$ and none uncleared. □

Theorem 4.4 (computer-assisted)**.** *Let* (G, ℓ) *be a leveled graph in which no two vertices of level* 1 *are adjacent. If* G *has at most five vertices, or* G *has six vertices of which at most two have level* 1*, then* (G, ℓ) *is good. In particular the leveled conjecture of Section* sec:leveled *of the paper holds for all graphs on at most five vertices.*

Proof. Goodness is preserved by passing to spanning subgraphs and by raising levels, so it suffices to treat $G = K_n - E(K_I)$ with $\ell = 1$ on an independent set I and $\ell = 2$ elsewhere, $|I| = s$. By Lemma 2.3, the leveled Lemma 2.1 ($D_v = \ell_v + 1$) and Corollary 2.2 it suffices to find colourings with $|g_v| \le \ell_v + 0.99$, and by Lemma 3.2 (with per-function thresholds) it suffices that no 2-sparse pattern with $j \le n$ items whose pairs are edges of G admits a bad configuration with thresholds $c_v = \ell_v + 0.99$. Patterns are enumerated up to the automorphisms $S_s \times S_{n-s}$ of (G, ℓ) (orbit-sum check passed for every (n, s, j) used). In (N3) only functions with equal thresholds are exchanged. For $n \le 5$, $1 \le s < n$: 3966 patterns, 11200 boxes; for $n = 6$, $s = 1$: 4755 patterns, 13810 boxes; for $n = 6$, $s = 2$: 7297 patterns, 37139 boxes; no uncleared box. The case $s = 0$ is Theorem thm:H(b) (resp. Theorem 4.1), and $s = n$ is trivial. Sanity checks: the method does *not* clear K_2 with levels $(1, 1)$ (which is not good), nor a single function at threshold 0.99 (as $\mathcal{K}_1 = 1$), nor K_2 at threshold 1.99; it clears K_2 with levels $(1, 2)$ (Lemma lem:K12lv). □

For $n = 6$ and $s \ge 3$ the corner method fails, for a reason that is again a limit of the residual method, not a counterexample (see Proposition 7.2).

Corollary 4.5. *A graph G on seven vertices is 2-good (every 2-sparse family with $H(\mathcal{F}) \subseteq G$ has discrepancy at most 2) unless its complement is a subgraph of C_4, of $3K_2$ or of K_3; that is, with the possible exception of K_7, $K_7 - e$, $K_7 - P_3$, $K_7 - 2K_2$, $K_7 - K_3$, $K_7 - P_4$, $K_7 - 3K_2$ and $K_7 - C_4$.*

Proof. Work with all levels 2. If some vertex has at most two neighbours, delete it by Lemma lem:cheap with $J = N(h)$; the remaining six-vertex graph with all levels 2 is good by Theorem 4.1 and Lemma lem:good. If some vertex has three neighbours, delete it with $|J| = 2$; the third neighbour drops to level 1 and Theorem 4.4 ($n = 6$, $s = 1$) applies. The remaining graphs have minimum degree at least 4, hence at most seven non-edges; for all graphs with at most seven non-edges (146 up to isomorphism) the program `seven_vertex_claude.py` searches all sequences of applications of Lemma lem:cheap (all h, all J) ending in a base of Theorem 4.4, and finds one for all but the eight listed graphs (`logs_claude/final/seven_vertex.log`). Goodness is inherited by subgraphs. □

Corollary 4.6. *Every 2-sparse family of eight functions has discrepancy at most 3 (the paper: 4).*

Proof. It suffices to show that K_8 with all levels 3 is good. Delete a vertex by Lemma lem:cheap with $|J| = 3$: the other four neighbours drop to level 2, leaving K_7 with levels $(2, 2, 2, 2, 3, 3, 3)$. Delete a vertex of level 2 with J consisting of two vertices of level 2: the remaining vertex of level 2 drops to 1 and the three of level 3 drop to 2, leaving K_6 with levels $(1, 2, 2, 2, 2, 2)$, which is good by Theorem 4.4 ($s = 1$). All applications of Lemma lem:cheap are legal: the deleted vertex has level $\geq 2 \geq |J|$ resp. $3 \geq |J|$, and every non-cheap neighbour has level ≥ 2. □

Theorem 4.7 (computer-assisted)**.** *$\mathcal{K}_5 = 3$ and $\mathcal{K}_6 = 4$. Moreover $\mathcal{K}_8 \geq 5$.*

Proof. Lower bounds. $\mathcal{K}_5 \geq \mathcal{K}_4 = 3$. For $\mathcal{K}_6 \geq 4$ take the six additive functions $g_i(x) = \langle a_i, x\rangle$ on six items with

$$a_1 = -\mathbf{1},\ a_2 = (1,1,-1,-1,-1,-1),\ a_3 = (1,-1,1,-1,-1,-1),\ a_4 = (-1,1,1,1,-1,-1),\ a_5 = (-1,1,1,-1,1,-$$

They are odd and 2-Lipschitz; $\langle a, x\rangle = 6 - 2\operatorname{dist}(x, a)$, so $|\langle a_i, x\rangle| \geq 4$ iff x is within distance 1 of a_i or $-a_i$, and the twelve balls of radius 1 around $\pm a_i$ cover $\{\pm 1\}^6$ (an antipodal covering code; exhaustive check: $\min_x \max_i |g_i(x)| = 4$). For $\mathcal{K}_8 \geq 5$ take the eight additive functions on seven items whose coefficient vectors are one word from each complementary pair of the Hamming $[7, 4]$ code (which contains $\mathbf{1}$): since the code is perfect of radius 1, every x is within distance 1 of some $\pm a_i$, whence $|\langle a_i, x\rangle| = 7 - 2\operatorname{dist}(x, \pm a_i) \geq 5$ (exhaustive check: the minimum is 5).

Upper bounds. By Corollary 2.2 with $D = 4$ (resp. $D = 5$) it suffices to show discrepancy at most 3.99 for five functions (resp. 4.99 for six), and by Lemma 3.2 (full pattern) that for $j \leq n$ no p admits a bad configuration for n functions reading all j items. All items are interchangeable, so (N2) allows $p_1 \leq \cdots \leq p_j$. The branch-and-bound clears $n = 5$, $c = 3.99$, $j = 1, \ldots, 5$ with $1, 1, 1, 3, 29$ boxes, and $n = 6$, $c = 4.99$, $j = 1, \ldots, 6$ with $1, 1, 1, 1, 3, 15$ boxes. The paper's Python checker gives the same results and box counts (with its own coordinate sorting), and our C program gives the same counts for $n = 5$ also without sorting. Sanity checks: our program reproduces the paper's run $\mathcal{K}_4 \leq 3$ at $c = 3$ (13 boxes for $j = 4$) and fails, as it must, for four functions at $c = 2.99$ and for six functions at $c = 3.99$ (near $p = (\frac{1}{2}, \ldots, \frac{1}{2})$, where the additive example above lives). □

This answers Open Problem 5 of the paper ("Is $\mathcal{K}_5 = 3$?"), and shows $\mathcal{K}_5 < \mathcal{K}_6$. Consequently the bound $\mathcal{K}_t$ of Theorem thm:A of the paper equals 3 for $t = 5$ and 4 for $t = 6$ (Berge-acyclic patterns). The known values are now $(\mathcal{K}_1, \ldots, \mathcal{K}_6) = (1, 2, 2, 3, 3, 4)$, with $\mathcal{K}_7 \geq 4$ (see Section 6) and $\mathcal{K}_8 \geq 5$.

5 An $O(\sqrt{n})$ bound for arbitrary bundle sizes

The following simple observation is not in the paper. It removes the logarithm from the bound of HMMS for sparse families, and, unlike Theorem thm:sqrtn of the paper, it does not depend on the number of items shared by two functions.

Theorem 5.1. *Let $t \geq 2$ and let $\mathcal{G}$ be a t-sparse instance with n functions. Then* $\operatorname{disc}(\mathcal{G}) \leq \sqrt{2\ln(2t)\,n}$. *In particular every 2-sparse family of n functions, with arbitrary bundle sizes and private items (for example on K_n), has* $\operatorname{disc} \leq \sqrt{2\ln 4}\,\sqrt{n} < 1.666\sqrt{n}$.

Proof. Let x, J ($j = |J| \leq n$), p and the restrictions h_v be as in the proof of Lemma 3.2, and let $d_v = |J_v|$ be the number of fractional items in U_v. Then $\sum_v d_v \leq tj \leq tn$, since every item lies in at most t supports. If $d_v = 0$, then h_v is the constant $G_v(x) = 0$. Let $Z \sim w(p)$. For $d_v \geq 1$, $h_v(Z)$ is a function of d_v independent coordinates with bounded differences 2 and mean 0, so by McDiarmid's inequality

$$\Pr\bigl[|h_v(Z)| \geq \lambda\bigr] \leq 2\exp\bigl(-\lambda^2/(2d_v)\bigr) \qquad (\lambda > 0).$$

Let $C^2 > 2\ln(2t)$ (so $C^2 > 2$) and $\lambda = C\sqrt{n}$. The function $\phi(d) = d^{-1}\exp(-C^2 n/(2d))$ satisfies $(\ln\phi)'(d) = (C^2 n/2 - d)/d^2 > 0$ for $0 < d \leq n$, so $\phi(d) \leq \phi(n)$, that is $\exp(-C^2 n/(2d)) \leq (d/n)\,e^{-C^2/2}$, for $1 \leq d \leq n$. Hence

$$\sum_v \Pr\bigl[|h_v(Z)| \geq C\sqrt{n}\bigr] \leq \frac{2}{n}\,e^{-C^2/2}\sum_v d_v \leq 2t\,e^{-C^2/2} < 1,$$

and some corner z has $|h_v(z)| < C\sqrt{n}$ for all v; with the integral coordinates of x this is a colouring of discrepancy $< C\sqrt{n}$. As $C^2 \downarrow 2\ln(2t)$, the minimum (over finitely many colourings) is at most $\sqrt{2\ln(2t)\,n}$. □

Remark 5.2. (0) The proof is a routine refinement of the union bound in the randomized rounding of HMMS (their Theorem 8); we found it neither in the paper nor (as far as we can tell from the paper's account of HMMS and of Dupré la Tour–Fujii) in those works, but its novelty should be checked against them before it is claimed beyond the paper. It supersedes Theorem thm:sqrtn(b) of the paper for every s, and Theorem thm:sqrtn(a) except for $s = 1$, where $1 + 2\lceil\sqrt{n/2}\,\rceil \approx 1.414\sqrt{n}$ is better. (1) For K_n with arbitrary bundles the previous best bounds were $\lfloor n/2 \rfloor$ (Corollary cor:maxtwod(iv)) and $\sqrt{2n(1+\ln 2n)}$ (HMMS); Theorem thm:sqrtn gives $1 + 2\lceil\sqrt{sn/2}\,\rceil$, which is better only when the multiplicities s are small. The HMMS union bound is over n functions that may each read n fractional items; sparsity limits the total number of (function, fractional item) incidences to tn, and the weighted union bound exploits exactly this. (2) By Proposition 7.1, on K_n with bundles of size at least n, no argument that uses only the facts listed there can give better than $a_n/2$, which is asymptotically $0.399\sqrt{n}$; so for such supports the residual method is determined up to a constant factor between about 0.399 and 1.666. (3) The bound is in terms of n, not of the degeneracy; it does not address Open Problem 2 of the paper ($o(d)$), except for patterns all of whose blocks are small: by Corollary cor:onesum, Theorem 5.1 and Lemma lem:good, if every block of $H(\mathcal{G})$ has at most N vertices then $\operatorname{disc}(\mathcal{G}) \leq \sqrt{2\ln 4\,N}$.

6 Status of the larger runs

All runs below use the same programs and conditions as above; "cleared" means: no uncleared box. None of the statements that would follow from them is claimed.

- *Seven* 2*-sparse functions* ($c = 2.99$): 4400 patterns ($2, 7, 23, 78, 264, 915, 3111$ for $j = 1, \dots, 7$; orbit sums checked). All patterns with $j \le 6$ are cleared (1801 boxes; the paper's Python checker, extended verbatim to local cubes of dimension 7, gives the same counts), and 3110 of the 3111 patterns with $j = 7$ are cleared (the paper's checker confirms 1866 of them with identical box counts). The remaining pattern is the dense pair "two functions sharing all seven items". Its branch-and-bound did not finish (the depth-first search is very slow near $p = (\frac{1}{2}, \dots, \frac{1}{2})$). At the single point $p = (\frac{1}{2}, \dots, \frac{1}{2})$ there is no bad configuration: by Remark 3.4 and Harper's vertex-isoperimetric theorem, a part U with $\mathbb{E}\,\mathrm{dist}(X, U) > 1.495$ has at most 29 points (initial segments of the simplicial order maximise $\mathbb{E}\,\mathrm{dist}(X, \cdot)$ among sets of a given size, and the segment of size 30 has $\mathbb{E}\,\mathrm{dist} = 1.4453$), and $4 \cdot 29 < 128$. Independently, the exact depth-first search of the program at this single point finds no cover (82,675,235 nodes, `logs_claude/pair7_uniform_point.log`). A third, independent check, an exact SAT encoding of the cover problem at this point (cardinality constraints, kissat), is UNSAT (`pair7_uniform_sat.log`). A sound version of this "capacity" test for boxes (with the envelope ratio $R = \max_y \mathrm{UB}(y) 2^d / \mathrm{DEN}^d$, `corner_bnb_v9_claude.c`, option `CAPACITY`) clears boxes only very close to the uniform point. Hence "every 2-sparse family of seven functions has discrepancy at most 2" (and K_7) remains *open* here.
- *Leveled six-vertex graphs*: cleared for $s \le 2$ level-1 vertices (Theorem 4.4); for $s \ge 3$ the method fails (Proposition 7.2).
- $\mathcal{K}_7$ ($c = 4.99$, full pattern): $j \le 6$ cleared; $j = 7$ did not finish. A clearance would give $\mathcal{K}_7 = 4$.
- *Eight* 2*-sparse functions* ($c = 2.99$, 256-bit corner sets, 128-bit arithmetic): 17652 patterns (12789 with $j = 8$, orbit sums checked); $j \le 6$ cleared; most patterns with $j = 7, 8$ did not finish (dense patterns are slow), none failed.

7 Limits of the residual method

Lemma 3.2 uses only three facts about the fractional point: at most n fractional items, the sparsity pattern of the fractional items, and that the restricted functions are 2-Lipschitz with mean zero under a product measure. The next proposition shows that no argument using only these facts can prove Conjecture conj:two for 25 or more functions, or give $o(\sqrt{n})$ for K_n.

Proposition 7.1. *Let $k \ge 1$, $S(z) = \sum_{i=1}^{k} z_i$ on $\{\pm 1\}^k$, let w be the uniform distribution and $a_k = \mathbb{E}_w|S|$. Then $h_1 = a_k - |S|$ and $h_2 = S$ are 2-Lipschitz with mean 0, and $\max(|h_1(z)|, |h_2(z)|) \ge a_k/2$ for every z. Moreover $\min_z \max(|h_1(z)|, |h_2(z)|) = 3$ for $k = 25$ (where $a_{25} = 4.0295\ldots$). Since $a_k = (1 + o(1))\sqrt{2k/\pi}$, the residual pattern "two functions sharing k items" has residual discrepancy $\Omega(\sqrt{k})$.*

Proof. Changing one coordinate changes S by 2 and $|S|$ by at most 2. The means are 0 by symmetry and by the choice of a_k. If $|h_2(z)| = |S(z)| = s$ then $|h_1(z)| = |a_k - s|$, and $\max(s, |a_k - s|) \ge a_k/2$. For $k = 25$, s is odd, and $\max(s, |a_{25} - s|)$ equals $3.03\ldots, 3, \ldots$ for $s = 1, 3$, and is ≥ 5 for $s \ge 5$ (exact rational computation in `logs_claude`). □

Consider supports with $n \ge 25$ functions in which two functions share a bundle of at least 25 items (e.g. K_n with $|B_{12}| \ge 25$). The pattern in which $j = 25 \le n$ fractional items all lie in that bundle is one of the patterns for which Lemma 3.2 requires that no bad configuration exists, and at p uniform the configuration of Proposition 7.1 is a bad configuration at every threshold $c < 3$. So the corner method cannot prove disc ≤ 2 for such supports. Likewise, at threshold c it fails as soon as $a_k/2 > c$ for some $k \le n$ not exceeding the largest bundle size, so on K_n with bundles of size

at least n it gives at best $\Theta(\sqrt{n})$ (asymptotically at best $(a_n/2) \approx 0.399\sqrt{n}$). At the paper's own threshold $c = 2$ the same configuration already blocks the method for $k \geq 15$ ($a_{15} = 6435/2048$; exact values in `logs_claude/final/pairbarrier_exact.log`, where $k = 25$ is also seen to be the least k with $\min_z \max(|h_1|, |h_2|) \geq 3$). We do not claim that these configurations are residuals of actual worst instances; the statements concern the method.

Proposition 7.2. *Let $n = 6$, $I = \{0, 1, 2\}$ with level 1 and $3, 4, 5$ with level 2, $G = K_6 - E(K_I)$. The 2-sparse pattern with $j = 6$ items in which items $2i, 2i + 1$ are read by i and by 3 ($i = 0, 1, 2$), and functions $4, 5$ read nothing, admits, at the uniform measure, the bad configuration $h_i(z) = z_{2i} + z_{2i+1}$ ($i \in I$), $h_3(z) = 3 - 2\left|\{i : z_{2i} = z_{2i+1}\}\right|$, $h_4 = h_5 = 0$, even for the exact thresholds $(1, 1, 1, 2, 2, 2)$: all are 2-Lipschitz with mean 0, and every corner has $|h_i| = 2$ for some $i \in I$ or $h_3 = 3$. Hence the residual method cannot prove that this leveled graph is good; the same pattern embeds in G for $s \in \{3, 4, 5\}$. The obstruction lives on the star $K_{1,3}$ with centre 3 at level 2 and leaves at level 1 (using $j = 6 > 4$ items, which is allowed because the other two functions read no fractional item); that star is good by Theorem* thm:leveled*(a) of the paper, so the barrier is purely methodological. It suggests combining the corner method with the leaf and cheap-deletion rules for $n = 6$, $s \geq 3$.*

Proof. Direct verification over the 64 corners (exact; `leveled_star_residual_check.log`). If all three pairs are mixed, $h_3 = 3$; otherwise some pair is constant and the corresponding $|h_i| = 2$. □

8 Negative findings on parallel transfers

We record what we tried towards a multi-hub transfer that would break the counting barrier (Remark rem:nearlimits(3)), and why it fails.

1. *Product-form joint demands reduce to sequential cheap deletions.* For hubs $h_1, \ldots, h_q$ deleted together, a demand on a common neighbour r of the form "the slice of r over $B_{h_1 r} \cup \cdots \cup B_{h_q r}$ contains a product (or an iterated fibration) of near-loops" is exactly what q successive applications of Corollary cor:neartransfer give at no cost to r, and each hub must then absorb one unit of deficiency per such neighbour; the example of Remark rem:nearlimits(2) applies to each hub. So such demands obey the counting barrier.

2. *A single joint loop is too weak.* The demand "the joint slice of r over $B_{h_1 r} \cup B_{h_2 r}$ contains a 1-loop" costs r one level for two edges (Lemma lem:hub with $S = B_{h_1 r} \cup B_{h_2 r}$), but it does not let both hubs succeed: with one common neighbour, the loop that first flips $B_{h_1 r}$ and then $B_{h_2 r}$, and $g_{h_i} = \sum_{B_{h_i r}} s$ ($|B_{h_i r}|$ odd, large), give $|g_{h_1}| \leq 1$ only while $B_{h_2 r} = \mathbf{1}$, where $|g_{h_2}| = |B_{h_2 r}|$. Conceptually, the joint configuration space of the hubs is then $\mathbb{Z}_2$-equivalent to a space of index 1 (two cubes glued along their corners), which cannot force two odd equations (the torus obstruction of Proposition prop:torus).

3. *Global $(\mathbb{Z}_2)^r$-equivariant configuration spaces on K_n.* For a group acting on bundle e through a character λ_e, a function g_v splits into $2^{D_v - 1}$ equivariant pieces, $D_v = \operatorname{rank}\{\lambda_e : e \ni v\}$; on K_n, small ranks force all bundles carrying the same character to carry the same (large) number of cuts, and large ranks give exponentially many pieces. We found no choice giving $O(1)$ fractional items per function.

4. *Residual (partial colouring) repair* after the HMMS point loses oddness; mean-zero is the only surviving property, and Proposition 7.1 bounds what it can give.

9 Files

All in `dense3_claude/`. Code (`code_claude/`):

- pattern enumeration: `patterns_claude.py` (brute-force canonisation, $n \leq 6$), `patterns_aug_claude.py` ($n \leq 7$) and `patterns_aug3_claude.py` ($n = 8$) (augmentation), `patterns_lev_claude.py` (leveled); all print orbit-sum checks;
- branch-and-bound: `corner_bnb_v7_claude.c` is the final source (per-function thresholds `LEVELS`, full-pattern mode `FULL`, threshold-aware symmetry breaking, (N2) sorting, incremental "alive vertex" feasibility tables); `corner_bnb_v8_claude.c` is the same with 256-bit corner sets ($n = 8$). Earlier versions v3 (uniform thresholds; used for the first $n = 6, 7$ runs) and v6 (= v7 without the incremental tables) give identical box and node counts on every file we compared; v4/v5 must not be used for leveled runs (no threshold-aware symmetry breaking);
- `run_final_claude.sh`: re-runs every computation behind a claimed result with the final source and writes logs with headers (date, source checksum, command) to `logs_claude/final/`;
- paper checker: `cover_bnb_paper_copy_claude.py` (verbatim copy of the paper's `cover_bnb.py`), drivers `indep_check_claude.py`, `indep_full_claude.py` (which extend its table of local distance matrices to dimensions $7, 8$ by the same formula);
- `run_lev_claude.sh`, `kn_schedule_bases_claude.py` (schedules of cheap deletions on K_n), `residual_milp_claude.py` (MILP for residual optima at a fixed p, exploratory only).

Logs (`logs_claude/`): `final/` (claimed results, final source, with headers; `paperchecker_*.log`; `pairbarrier_exact.log`); `n6c299/`, `indep/`, `lev/`, `Kgen/` (original runs, including the lower-bound searches `K6_lower_bound_search.log`, `K8_hamming_lower_bound.log`), `leveled_star_residual_check.log`, `sqrtn_inequality_check.log`; `n7*`, `n8*`, `indep7/` (runs for Section 6). The referee's independent checker and logs are in `referee_claude/`.